DATA SCIENCE MODULE PROJECT · TEAM 2

S&P 500 Stock Data Pipeline & Performance Analysis

End-to-end ELT engineering, automated data quality testing, and market return analytics across 600k+ stock records

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Coverage: 622,710 records · 503 Tickers · 2021–2026

EXECUTIVE SUMMARY

Central business question: With Data Cleaning techniques, can accurate and precise analysis be performed on free and readily available public data to arrive at useful insights for everyday investors?

Robust ELT Architecture

Extracted 622k+ rows of static S&P 500 daily price and fundamental data.

Loaded from Supabase PostgreSQL to Google BigQuery via Meltano ELT.

Transformed into high-performance star schema models using dbt.

Core Business Insights

Stock-level insight

AAPL Return Growth: No incremental annual return growth since the 2023–2024 period.

NVDA Sharpe Ratio: Exceptional risk-adjusted Sharpe ratio > 1 over 3- and 5-year horizons.

Sector Risk-Reward: Tech led in daily return (+0.21%) but exhibited greatest volatility (3.60%).

Price-profits: Tech earned its rally (+30% price +32% profits) vs. Financials grew profits without moving price (+1%, +23%)

Price momentum

Price vs. Moving Average to show if stock is on uptrend, (short-term or deeper) pullback, or downtrend.

INFRASTRUCTURE STRATEGY

Technical Approach: Tool Selection Rationale

Google BigQuery

Serverless scale for 600k+ rows and 38 fields using GCP credits.

Supabase & Meltano

PostgreSQL staging with automated CLI ELT to cloud warehouse.

dbt Modeling

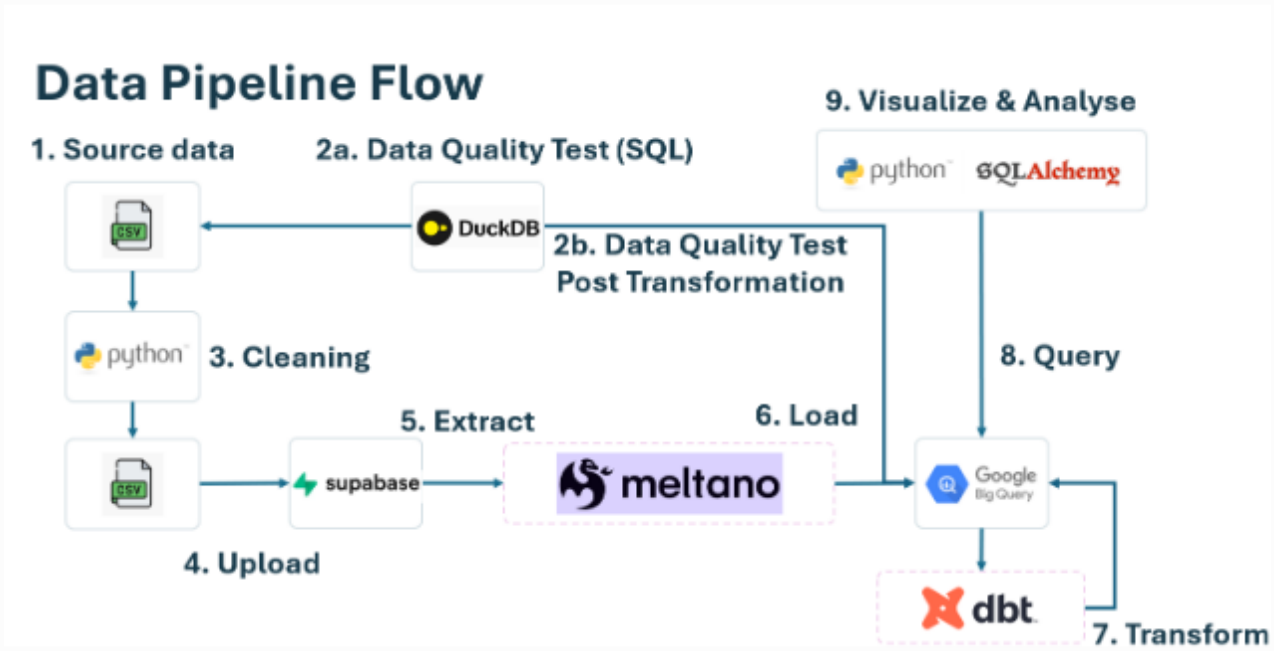
Transforms raw tables into star schema with automated test suites.

Python & SQLAlchemy

Direct connection to BigQuery for vectorized Pandas analysis.

END-TO-END WORKFLOW

Data Pipeline Architecture & Execution Stages



1. Source & Extraction

503 S&P 500 tickers, Yahoo Finance prices, SEC EDGAR XBRL.

2. Staging & Meltano ELT

Loaded to Supabase, piped via tap-postgres to target-bigquery.

3. Transform & Quality Test

dbt star schema transformation and automated integrity tests.

4. Analysis Layer

SQLAlchemy querying directly into Pandas for financial EDA.

DATA ENGINEERING RIGOR

Data Quality Testing & Anomaly Remediation

⚠ Initial V1 SQL Findings

Waters (WAT): ~1,000× share-count error causing $43T market cap.

McDonald's (MCD) share collapse and Halliburton (HAL) EPS distortions.

✓ Cleaned V2 Regression Results

Corrected shares × EPS cross-checks against net income.

Zero duplicate records, zero orphan facts, and zero return mismatches.

622,710	503	0	100%
Stock-Day Records	Validated Tickers	Duplicate Key Records	Daily Return Reconciliation

Data quality testing framework

A layered control set covering source integrity, business rules, analytical consistency, and warehouse validation.

SOURCE & RECORD CONTROLS

Source integrity	File hash, row count, column count, date range
Schema validity	Dates, numeric prices, integer-like fields
Completeness	NULLs and blank key fields
Uniqueness	One ticker + date record
Price validity	OHLC relationships; prices > 0
Volume validity	Non-negative volume
Cross-field reconciliation	Market cap vs price × shares

ANALYTICAL & WAREHOUSE

Financial reasonableness	Implausible P/E, EPS, shares, market cap
Outliers	Extreme values and scale errors
Temporal consistency	Example: WAT shares ~59m → ~59bn (≈1,000×)
Snapshot consistency	Example: compare every company on 2026-10-06
Coverage / population	491 at start + 12 later entrants = 503 latest
Referential integrity	Every fact ticker must exist in dim_ticker (0 missing)
Post-ELT validation	Source preservation + derived-field checks

Test evidence matrix

Every conclusion is traceable to a SQL script and a corresponding CSV result file in the project repository.

#	Test	Why	SQL	Result evidence	Outcome
01	**Baseline / source integrity**	Establish control totals	01_baseline.sql	01_baseline.csv	PASS
02	**Completeness**	Quantify missing fields	02_completeness.sql	02_completeness.csv	REVIEW
03	**Uniqueness**	Prevent duplicate stock-day rows	03_duplicates.sql	03_duplicates_summary.csv	PASS
04	**Price & volume validity**	Validate OHLCV business rules	04_price_business_rules.sql	04_price_business_rules.csv	PASS
05	**Cross-field integrity**	Detect financial scaling inconsistencies	05_financial_integrity.sql	05_financial_integrity.csv	REVIEW
06	**Reasonableness / outliers**	Flag implausible financial values	06_financial_reasonableness.sql	06_financial_reasonableness.csv	REVIEW
07	**Temporal consistency**	Detect step-change scale errors	07_temporal_consistency.sql	07_temporal_consistency.csv	REVIEW
08	**Snapshot / coverage**	Control date and constituent population	08_snapshot_coverage.sql	08_snapshot_coverage.csv	REVIEW
09	**Referential integrity**	Validate fact-to-dimension keys	09_bigquery_referential_integrity.sql	bigquery/09_...csv	PASS
10	**Post-ELT validation**	Reconcile source and derived fields	10_post_elt_validation.sql	bigquery/10_...csv	PASS

Submission pack: SQL scripts ZIP + CSV results ZIP. Each matrix row maps directly to its evidence file in the ZIP.

Exceptions worth presenting

The controls exposed material scaling anomalies that simple reconciliation could miss.

WAT • Waters

~1,000× share-count scale anomaly

~59m shares appeared as ~59bn, creating an implausible >$10T market cap.

MCD • McDonald's

Share count collapsed to hundreds

Hundreds of millions of shares temporarily appeared as hundreds before reverting.

HAL • Halliburton

Extreme EPS scaling anomaly

TTM EPS jumped from ordinary values to an extreme ~100,001.63 level.

BE • Bloom Energy

Quarterly EPS temporal anomaly

Diluted EPS exhibited ≥1,000× period-to-period movement requiring review.

KEY CONTROL INSIGHT

Two wrong fields can still reconcile.

A market-cap check alone can pass if market cap and shares outstanding carry the same upstream scale error. Independent temporal and reasonableness tests are therefore essential.

93 temporal anomaly events across 12 tickers — counts below sum to 93



Detailed exception rows: 05_share_count_exceptions.csv • 06_financial_outlier_exceptions.csv • 07_temporal_exceptions.csv

S&P 500 membership changed during the five-year dataset

491 of the 503 latest tickers were already present on the first date; 12 entered later. This is a population change, not automatically a data error.

491
securities on first observation date

+12 entrants

503
securities on latest observation date

11
sectors

0
exits before latest date

How to read 491 + 12 = 503

- 491 tickers are present from the first observation date through the latest date.
- 12 tickers entered the dataset later, giving 503 on the latest date.
- For rankings or comparisons, use one common date (or a fixed cohort) so the comparison is fair.

Practical rule

Same-date snapshot or fixed cohort = fair comparison

A later entrant should not be treated as “missing historical data” simply because it was not in the starting S&P 500 population.

Post-ELT validation: warehouse transformation passed

PASS

How we checked it: row/ticker counts, ticker-date keys, fact-to-dimension joins, plus independent recalculation of SMA and adjusted-close daily returns.

Raw source
public_proj2_sp500
622,689 rows **503 tickers**

ELT

Fact table
fact_dailyprice
622,689 rows **503 tickers**

JOIN

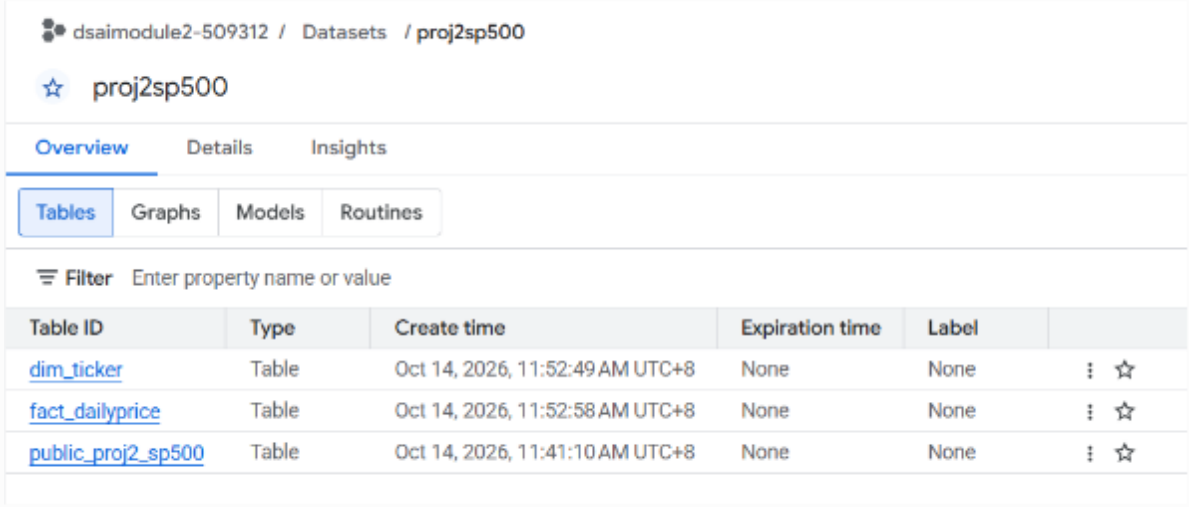
Dimension
dim_ticker
503 rows **503 unique tickers**

0	source keys lost
0	unexpected fact keys
0	orphan fact rows
0	duplicate ticker-date rows
0	SMA mismatches
622,186 / 622,186	daily returns reconciled

Conclusion • Transformation integrity passed: no keys were lost or added, joins were complete, and derived fields reconciled. PASS here does not mean every upstream source anomaly was cleaned; those exceptions remain separately documented for remediation.

DATA MODELING

Star Schema Transformation via dbt in BigQuery



dim_ticker (Dimension Table)

Stores ticker, company name, GICS sector, and sub-industry mappings.

fact_dailyprice (Fact Table)

Stores daily OHLCV prices, market cap, and valuation multiples.

Derived Technical Indicators

Precomputes SMA 20, 50, 100, 200, and adjusted daily return fields.

ANALYTICS INTEGRATION

Querying BigQuery with Python & SQLAlchemy

```
DataFrame Output:
  ticker        date   close    sma_20    sma_50  daily_return
0   AAPL  2026-10-06  333.63  333.4115  322.3480      0.002223
1   AAPL  2026-10-05  332.89  332.5410  322.4136     -0.002397
2   AAPL  2026-10-02  333.69  331.8950  322.4162      0.010202
3   AAPL  2026-10-01  330.32  331.6210  322.1756     -0.008108
4   AAPL  2026-09-30  333.02  331.3530  322.0870      0.010990
5   AAPL  2026-09-29  329.40  330.9585  321.9814     -0.026596
6   AAPL  2026-09-28  338.40  330.3310  321.9252     -0.007828
7   AAPL  2026-09-25  341.07  329.3960  321.8320      0.015331
8   AAPL  2026-09-24  335.92  328.0715  321.6758     -0.003264
9   AAPL  2026-09-23  337.02  326.9480  321.5074     -0.008035

DataFrame Info:
ticker          object
date            object
close          float64
sma_20         float64
sma_50         float64
daily_return   float64
dtype: object
```

Secure Credentials

Uses python-dotenv and GCP credentials for programmatic authentication.

Optimized Star Joins

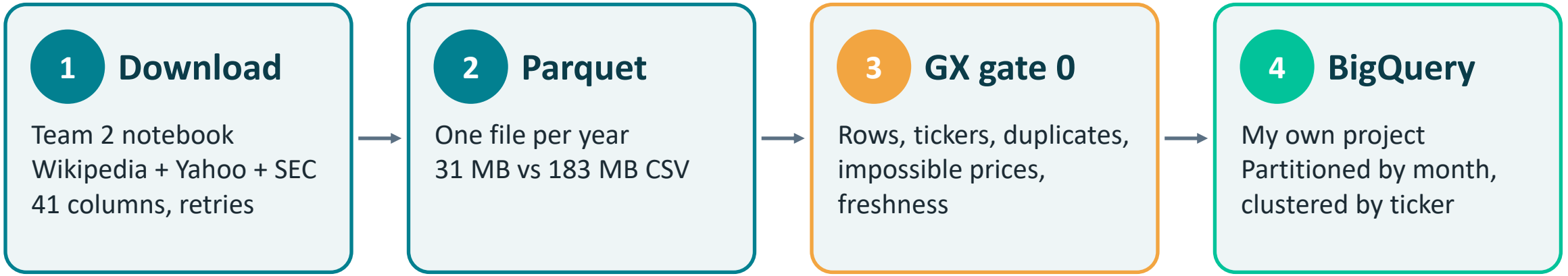
Joins fact_dailyprice with dim_ticker on ticker symbol efficiently.

Fast Pandas Ingestion

Streams schema-typed data into memory for statistical calculations.

A scheduled pipeline that stops itself when data is wrong

Tue–Sat at 6:30 am SGT, after each US trading day – no manual steps



Fails → retry once after 5 min → still failing: nothing is loaded, yesterday's table stays

622,757
rows loaded

503
S&P 500 stocks

14 Oct
last complete day

22 min
end to end

First run, 15 Oct 2026 · Dagster job own_v2_extract_and_load · download notebook by Team 2

Five quality gates – gate 3 caught a partial trading day

Gate 0
Download files
Before load

Gate 1
Raw table
Meltano load

Gate 2
Schema
4 + 41 columns

Gate 3
14 dimensions
Team 2 rules

Gate 4
Marts + views
Before analysis

8 Oct was downloaded while the US market was still trading

Closes that differ from my own (final) download



20

impossible prices flagged by gate 3 – open outside the day's high–low

228 → 0

Waters market caps above $10 tn, v1 → v2

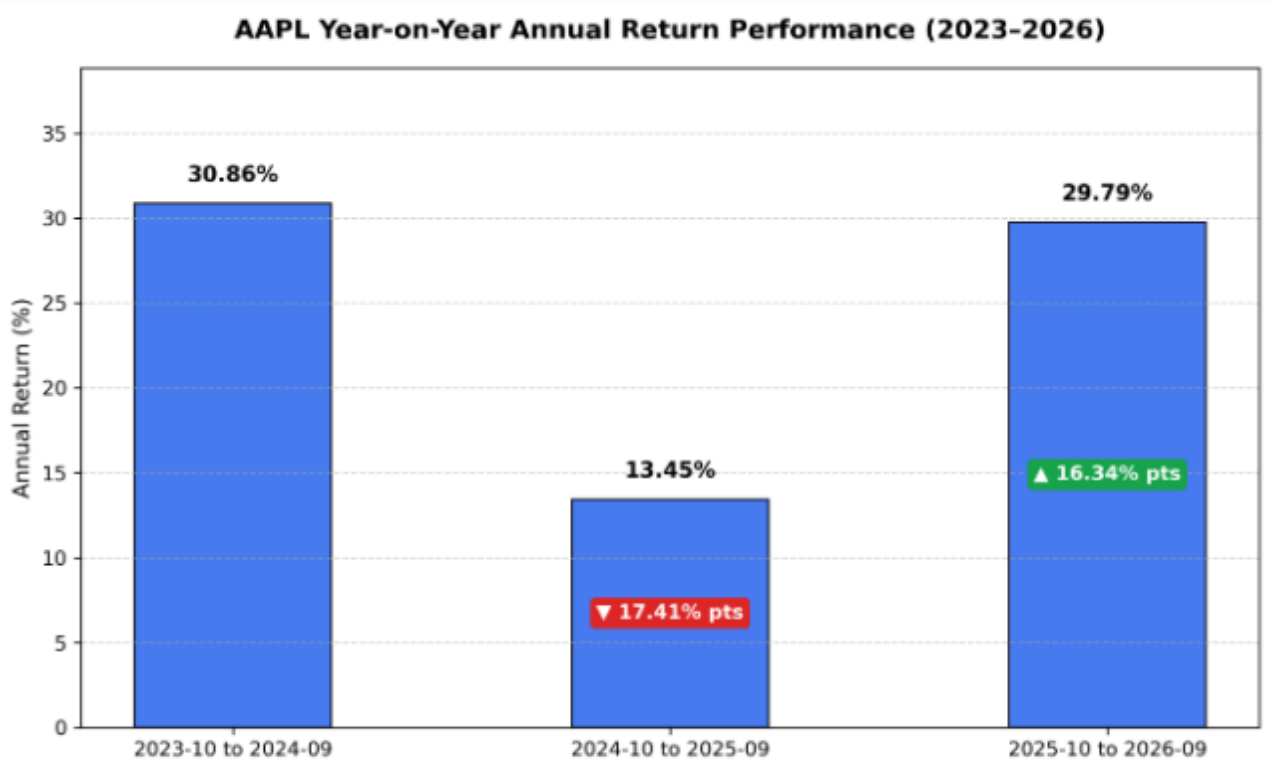
Action: analysis ends 7 Oct; evidence committed to the repo

Next: incremental loads · Docker + Google Cloud · project ID in .env

ANALYSIS 1 · AAPL TRAILING RETURNS

Is there growth in Apple Inc. annual return in last three 12 months trailing period?

Key Finding: No incremental growth on annual return since 2 years ago (since 202310–202409 period).



2 years ago: 2023-10 to 2024-09

+30.86%

Strong baseline performance year.

1 year ago: 2024-10 to 2025-09

+13.45%

17.41% points sharp deceleration.

latest: 2025-10 to 2026-09

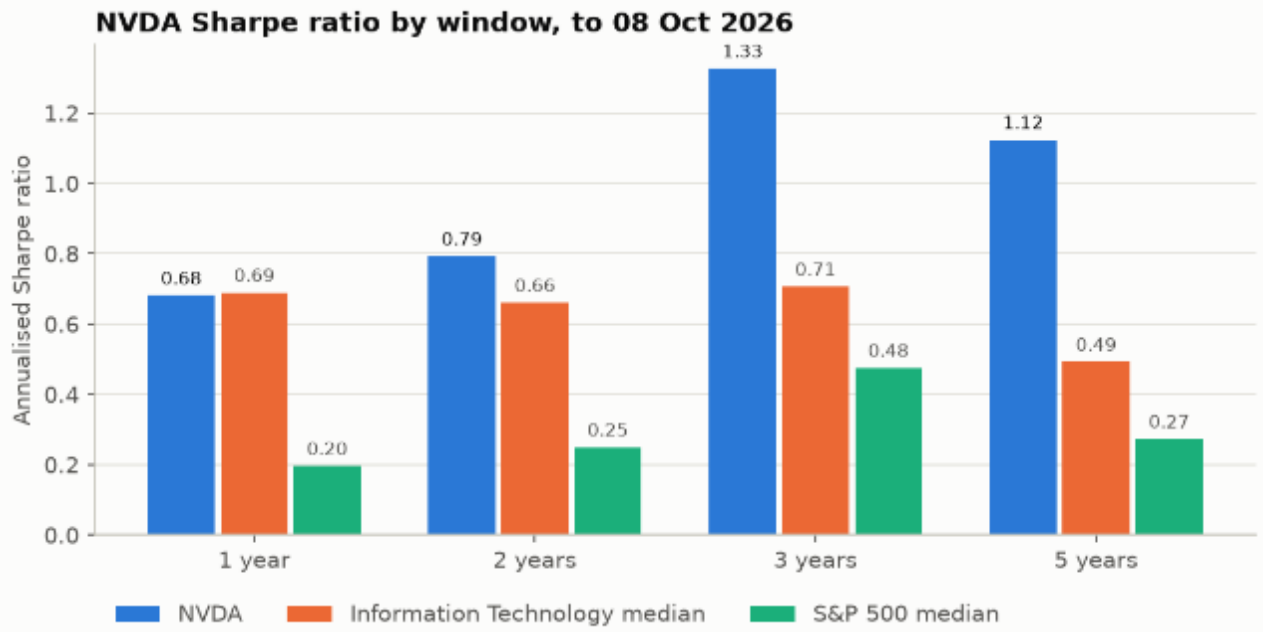
+29.79%

Rebound remains below 2024 peak level.

ANALYSIS 3 · RISK-ADJUSTED LEADERSHIP

Nvidia (NVDA): Multi-Window Sharpe Ratio Report

Key Finding: NVDA maintained exceptional Sharpe > 1 over 3 and 5 year horizons.



3-Year Window (AI Wave)

Sharpe: 1.33

95th percentile in S&P 500 (417.5% return).

5-Year Window

Sharpe: 1.12

98th percentile in S&P 500 (1038.4% return).

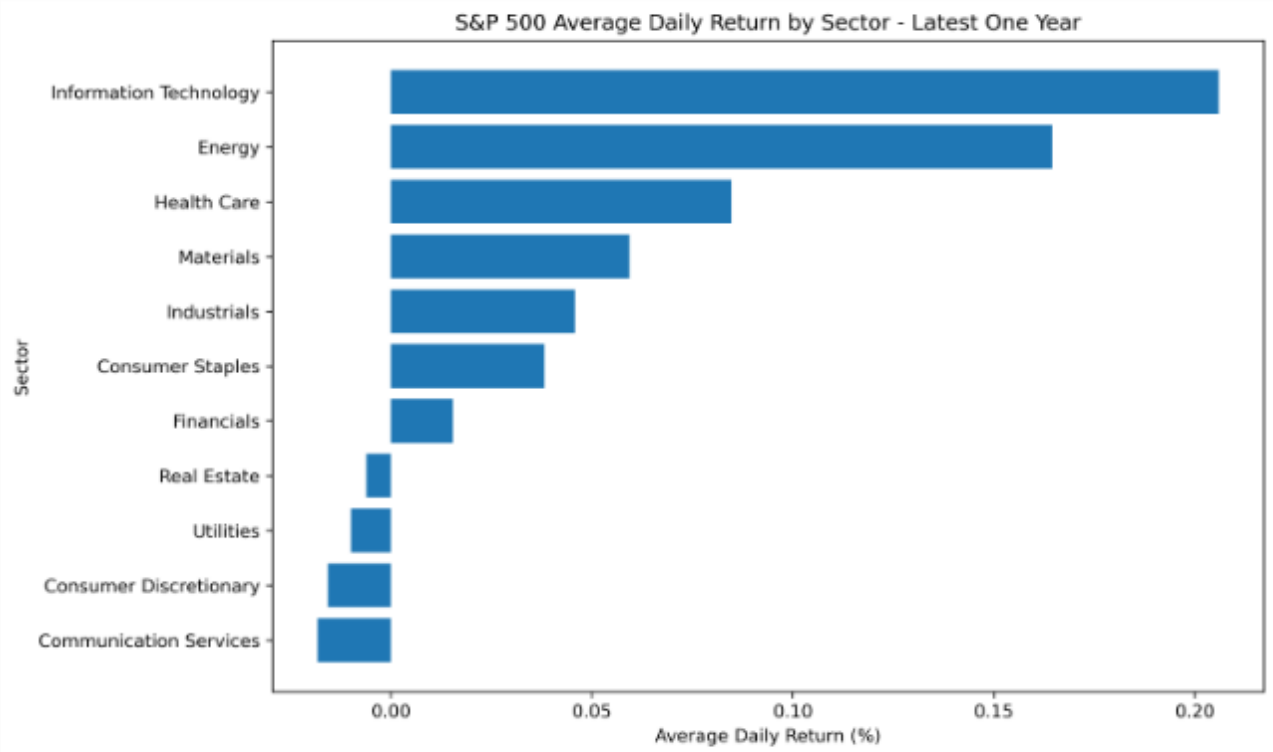
Above-Market Compensation

Sharpe > 1 proves excess returns comprehensively surpassed elevated volatility.

ANALYSIS 2 · SECTOR COMPARISON

S&P 500 Average Daily Return by Sector (Latest 1-Year)

Key Finding: Tech (+0.21%) and Energy (+0.16%) led all sectors in daily returns.



Top Performers
Info Tech: +0.2060%
Energy: +0.1645%

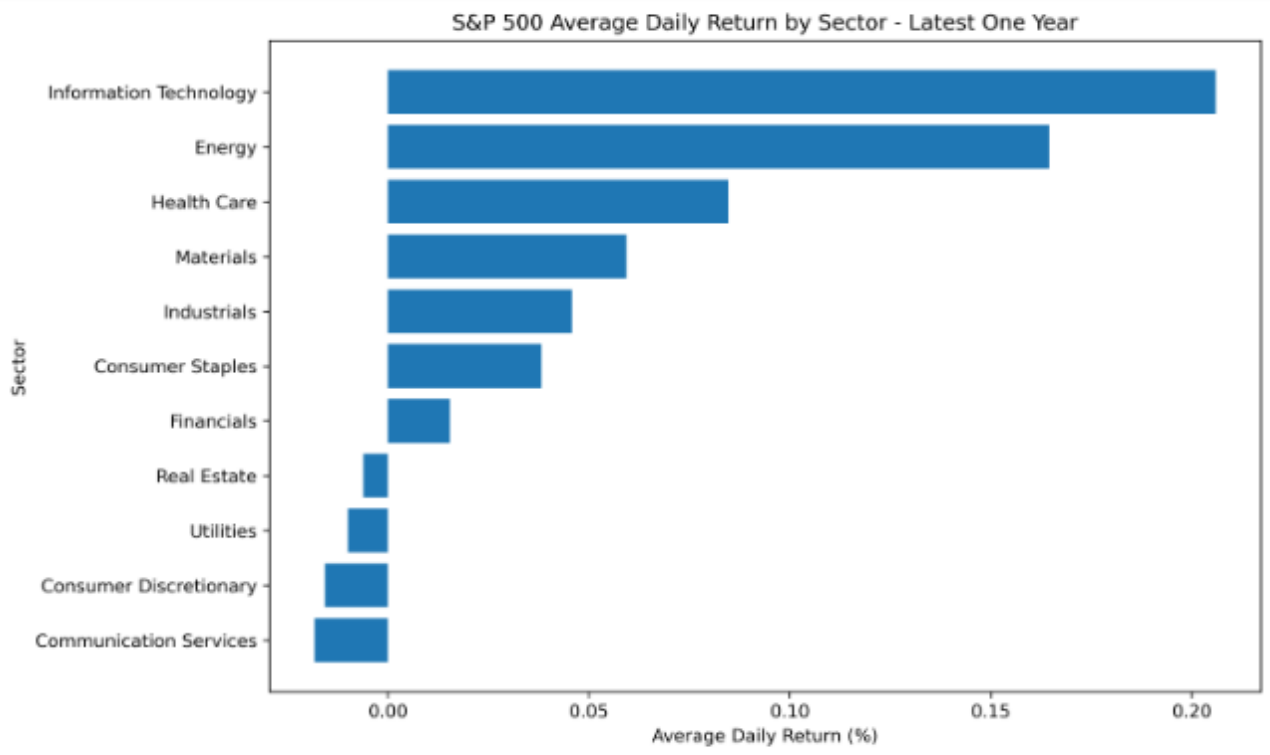
Positive Breadth
7 of 11 sectors posted positive average daily gains.

Lagging Sectors
Communication Services: -0.0182%
Consumer Discretionary: -0.0151%

ANALYSIS 2 · RISK VS. RETURN TRADE-OFF

Daily Return Volatility Across S&P 500 Sectors

Key Finding: Tech had highest returns but also the greatest daily variability (3.60%).



Highest Volatility
Info Tech: 3.6026% daily standard deviation.

Lowest Volatility
Utilities: 1.5530% daily standard deviation.

Risk-Return Principle
Superior tech performance required tolerating over 2.3× the daily price swings of utilities.

ANALYSIS 4 · SECTOR VALUATION

Did Prices Follow Profits? Price vs Earnings Growth by Sector

Key Finding: IT earned its rally; Financials grew profits 23% while prices stood still.



Information Technology

Price +30% · Profits +32%

Earned rally, but priciest sector (P/E 37.4).

Financials

Price +1% · Profits +23%

Lowest P/E of all sectors (14.3).

Method & Data Quality

Sector medians, 7 Oct 2025 – 7 Oct 2026 (8 Oct excluded). Real Estate omitted: distorted EPS.

ANALYSIS 5 · PRICE MOMENTUM

Example 1: Stock on the up (AAPL)

Data: BigQuery dsa1module2-509312.proj2sp500 • Updated daily

Ticker	Sector	Sub-Industry		Market Cap
AAPL	Informat…	Technology Hardware, Storage & Peripherals	Apple Inc.	$4933.30B



Price	SMA 20	SMA 50	SMA 100
$337.69	$335.03	$322.27	$315.43
	↑ $2.66	↑ $15.42	↑ $22.26

Strong uptrend
Price > MA20 >MA50 >MA 100

In past week, price came down slightly to intersect MA20, but still comfortably above MA50 and MA100

Above-Market Compensation
Stock in healthy, powerful bull market

ANALYSIS 5 · PRICE

Example 2: Short-term pullback (MSFT)

..

Data: BigQuery dsa1module2-509312.proj2sp500 • Updated daily

Ticker	Sector	Sub-Industry	Microsoft	Market Cap
MSFT	Informat...	Systems Software		$3933.67B



Price	SMA 20	SMA 50	SMA 100
$529.64	$508.48	$498.91	$448.41
	↑ $21.17	↑ $30.74	↑ $81.24

Short-term pullback
MA20 > Price in Sep

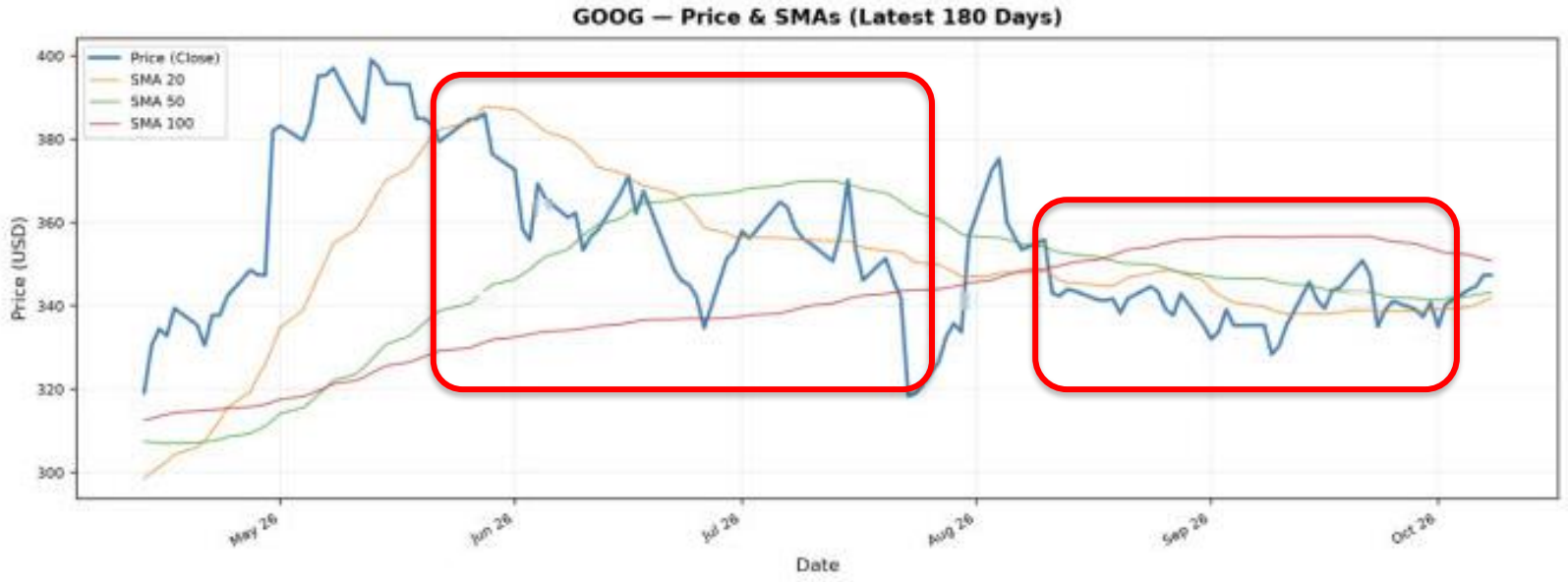
.
Brief cooling off period before buyers stepped back in in early Oct

ANALYSIS 5 · PRICE

Example 3: Deeper pullback (GOOG)

..

Ticker	Sector	Sub-Industry		Market Cap
GOOG	Commu…	Interactive Media & Services	Alphabet Inc. (Class C)	$4248.22B



Price	SMA 20	SMA 50	SMA 100
$347.36	$341.95	$343.16	$350.81
	↑ $5.41	↑ $4.20	↑ $-3.45

June pullback
Price below MA20, MA50 after Q2 earnings which rattled the market on capex

Sep pullback working through long term resistance
MA100 crosses price

Some uptick in early Oct
Is the stock ready to resume broader uptrend?

ANALYSIS 5 · PRICE

Example 3: Deeper pullback (GOOG) – zoom to past 360 days

..

Ticker	Sector	Sub-Industry		Market Cap
GOOG	Commu…	Interactive Media & Services	Alphabet Inc. (Class C)	$4248.22B



GOOG still on long-term upward trend

Pullback since
May (Price slipped below MA20)
Jun (MA50)
Aug (MA100)

Price	SMA 20	SMA 50	SMA 100
$347.36	$341.95	$343.16	$350.81
	↑ $5.41	↑ $4.20	↑ $-3.45

ANALYSIS 5 · PRICE

Example 4: Stock on structural downward trend

..

Ticker	Sector	Sub-Industry		Market Cap
NKE	Consum...	Apparel, Accessories & Luxury Goods	Nike, Inc.	$50.71B



Price	SMA 20	SMA 50	SMA 100
$34.18	$35.57	$38.29	$40.92
	↑ $-1.39	↑ $-4.11	↑ $-6.74

Price below all major MA
Price < MA20 < MA50 < MA100

.
Signals structural headwinds or fundamental business challenges

PROJECT SUMMARY

Summary of Engineering & Business Findings

1. Pipeline Reliability

Multi-layer validation resolved upstream financial anomalies and ensured clean cloud data tables.

2. Risk-Return Dynamic

Sector returns must be assessed with volatility; tech delivered top returns with 2.3× risk.

3. Alpha Leadership

NVDA demonstrated sustained risk-adjusted alpha (Sharpe > 1.3), outpacing peers across cycles.

Thank You & Questions